Template

Studentnames and studentnumbers here

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Set-up your environment

Title Page

Noel Kona Giorgia Tanzi Iann Soffried Julian Trinkl Morris Nouwens Dania Al Khulaifi

Part 1 - Identify a Social Problem

The social problem being discussed is whether the growth of wages keeps up with the rising costs of living. The central question of this research is: does the growth of wages of the lowest-paid workers rise enough for them to be able to afford their current lifestyle? And does that apply across Europe?

1.1 Describe the Social Problem

- The social problem: The social problem is about the prices of living rising faster compared to wages, resulting in a worker's real purchasing power falling even if their wages rise on paper. This leads to workers being able to afford less each year.
- The relevance: The problem of the growing wages and the rising cost of living is relevant, because in recent years many countries in Europe have faced a sharp rise in prices, while the growth of wages falls behind, leaving many households with low-income struggling to financially keep up with the new living costs (ILO, 2022). The EU also implemented Directive (EU) 2022/2041 on adequate minimum wages in 2022, this aims to keep wage levels keep up with the growing inflation and living costs (European Parliament and Council of European Union, 2022).
- What the report will answer: The report will fill in the gap by comparing consumer-price inflation and minimum-wage levels across European countries throughout recent years. Existing reports look globally/nationally, but cross-country researches are under-examined, which is why this report will use cross-country methods to deepen such

Part 2 - Data Sourcing

2.1: Data sourcing Eurostat was selected as the data source for both European Union's Harmonized Index of Consumer Prices (HICP) and the European Union's minimum wage. Eurostat uses a harmonised and standardised way of collecting statistics which allow for reliable cross-country comparisons. It collects and compiles data from national statistical offices using common methods to improve reliability and consistency. This makes Eurostat very suitable for determining whether minimum wages relative to the cost of living have been increasing at an adequate rate throughout Europe. There are some limitations to our data that deserve mention. The HICP data only takes into account average price movements of all consumer goods in a country and does not provide data for regional price variations or for the individual consumer's experience. Furthermore, the minimum wage database only contains data from countries that have followed a statutory National Minimum Wage system. This limits the comparisons between countries that have relied primarily on collective bargaining systems. Lastly, there were missing values for some observations, which resulted in an inability to include several countries in the comparison.

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

```
## # A tibble: 6 x 25
##   Data extracted on 09/0~1 ...2  ...3  ...4  ...5  ...6  ...7  ...8  ...9  ...10
##   <chr>                    <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 Dataset:                 Mini~ <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 2 Last updated:            30/0~ <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 3 <NA>                     <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 4 Time frequency           <NA>  Half~ <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 5 Currency                 <NA>  Euro  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 6 <NA>                     <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## # i abbreviated name: 1: `Data extracted on 09/06/2026 14:27:33 from [ESTAT]`
## # i 15 more variables: ...11 <chr>, ...12 <chr>, ...13 <chr>, ...14 <chr>,
## #   ...15 <chr>, ...16 <chr>, ...17 <chr>, ...18 <chr>, ...19 <chr>,
## #   ...20 <chr>, ...21 <chr>, ...22 <chr>, ...23 <chr>, ...24 <chr>,
## #   ...25 <chr>
```

```
## # A tibble: 6 x 33
##   Data extracted on 08/0~1 ...2  ...3  ...4  ...5  ...6  ...7  ...8  ...9  ...10
##   <chr>                    <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 Dataset:                 HICP~ <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 2 Last updated:            06/0~ <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 3 <NA>                     <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 4 Time frequency           <NA>  Annu~ <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 5 Unit of measure          <NA>  Annu~ <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## 6 Classification of indiv~ <NA>  Actu~ <NA>  <NA>  <NA>  <NA>  <NA>  <NA>  <NA>
## # i abbreviated name: 1: `Data extracted on 08/06/2026 14:18:56 from [ESTAT]`
## # i 23 more variables: ...11 <chr>, ...12 <chr>, ...13 <chr>, ...14 <chr>,
## #   ...15 <chr>, ...16 <chr>, ...17 <chr>, ...18 <chr>, ...19 <chr>,
## #   ...20 <chr>, ...21 <chr>, ...22 <chr>, ...23 <chr>, ...24 <chr>,
## #   ...25 <chr>, ...26 <chr>, ...27 <chr>, ...28 <chr>, ...29 <chr>,
## #   ...30 <chr>, ...31 <chr>, ...32 <chr>, ...33 <chr>
```

```
## Simple feature collection with 6 features and 11 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 1.442566 ymin: -13.43119 xmax: 74.88986 ymax: 42.6412
```

```
## Geodetic CRS:  WGS 84
## # A tibble: 6 x 12
##   CNTR_ID CNTR_NAME          NAME_ENGL NAME_FREN ISO3_CODE SVRG_UN CAPT  EU_STAT
##   <chr>   <chr>              <chr>     <chr>     <chr>     <chr>   <chr> <chr>
## 1 CD      République Démocr~ Democrat~ Républiq~ COD       UN Mem~ Kins~ F
## 2 CF      République Centra~ Central ~ Républiq~ CAF       UN Mem~ Bang~ F
## 3 CG      Congo-Kongo-Kongó  Congo     Congo     COG       UN Mem~ Braz~ F
## 4 AD      Andorra            Andorra   Andorre   AND       UN Mem~ Ando~ F
## 5 AE                  ~ United A~ Émirats ~ ARE       UN Mem~ Abu ~ F
## 6 AF            ~    - Afghanis~ Afghanis~ AFG       UN Mem~ Kabul F
## # i 4 more variables: EFTA_STAT <chr>, CC_STAT <chr>, NAME_GERM <chr>,
## #   geometry <MULTIPOLYGON [°]>
```

```
inline_code = TRUE
```

This study's aim is to assess wages' rise in line with the country's rising living costs across Europe through the use of three datasets.

The Harmonised Index of Consumer Prices (HICP) was used for tracking rent/housing rates in Europe. The data were sourced from Eurostat and cover the annual rate of inflation for Europe up to the present (2022). The dataset also enables us to analyze the way that housing costs change over time (i.e., rent inflation rates from 2015 – 2025).

The second dataset used is based on the growth of the minimum wage rate in European countries as supplied by Eurostat. The statistical measures capture the annual increase in the statutory minimum wage for the countries in Europe that have a National Minimum Wage. This data allows for comparison of wage increases with others and helps determine if wage growth has been enough to keep up with price increases.

The third dataset used contains the geographical boundaries for European Countries that were supplied through the spatial package 'giscoR'. These geographical boundaries are used to illustrate rent inflation differences between the countries of Europe with maps.

When combined, these three individual datasets allow for comparison of how, both over time the cost to rent housing and minimum wage are related.

2.3 Describe the type of variables included

Think of things like:

- Do the variables contain health information or SES information?

- Have they been measured by interviewing individuals or is the data coming from administrative sources?

Socioeconomic and economic indices, measured at the nation level, are the primary components of the datasets.

The rent data set comprises quantitative variables representing HICP (harmonised indices of consumer prices) values per year, which measure the change in consumer prices for housing-related items. This information allows analysts to assess how living costs have changed and how inflation is affecting households.

Wage datasets consist of quantitative variables that indicate how statutory minimum wage rates have changed from one year to another across the various countries. Labour force conditions and income levels among low-wage employees can therefore be analysed based on the available wage indices.

Country boundaries or country identifiers constitute the variables in the geographic dataset. While these geographic variables do not directly measure social or economic conditions, they allow for the geographical visualisation of socioeconomic indices.

The datasets originate from governmental administrations and official statistics rather than personal surveys or interviews. Economic data were gathered and developed by national statistical agencies and published by Eurostat. Thus, the datasets include standardised and equivalent figures across all European countries. we removed bosnia, turkey, and albania because some data was missing from them.

Variable	Type	Meaning
TIME	character	European country
year	integer	Year of observation
hicp	numeric	Harmonised Index of Consumer Prices
geometry	spatial	Country geographic boundaries

Part 3 - Quantifying

3.1 Data cleaning

How did we clean the data?

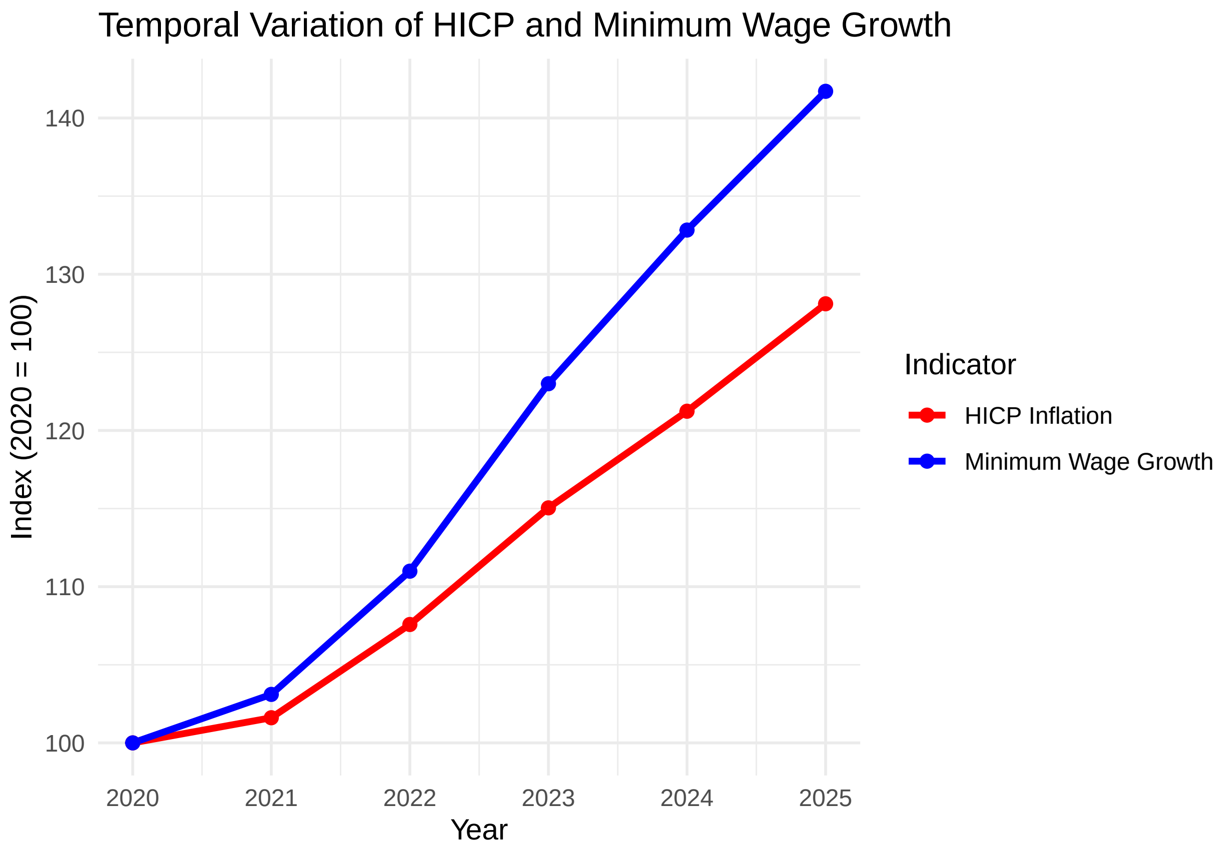
We processed the HICP, minimum wage, and geographic dataset. This involved a three-step procedure: 1. Cleaning of the data by removing metadata, unnecessary data rows, and unnecessary data columns: The original Eurostat datasets included descriptive metadata, empty or missing values, and variable definitions that were not needed for our research question; we did not require these records, so they were eliminated. Therefore, we retained only those country-level observations and years that were relevant to our analysis. 2. Standardizing and transforming the data into a more consistent format: We changed the format of both HICP and minimum wage values to a numeric format, and restructured each dataset in a consistent country-year format; Additionally, we also created a new variable that reflects the percent change in HICP between 2015 and 2025. This will help in comparing inflation trends between countries. 3. Merging datasets and removing incomplete observations: The cleaned HICP and minimum wage datasets were merged based on country and year matching records; In addition, the geographic dataset was merged based on matching records using country names. Any countries that did not have sufficient records to perform consistent and comparable analyses or visualizations were excluded.

3.2 Generate necessary variables

Creating new variables

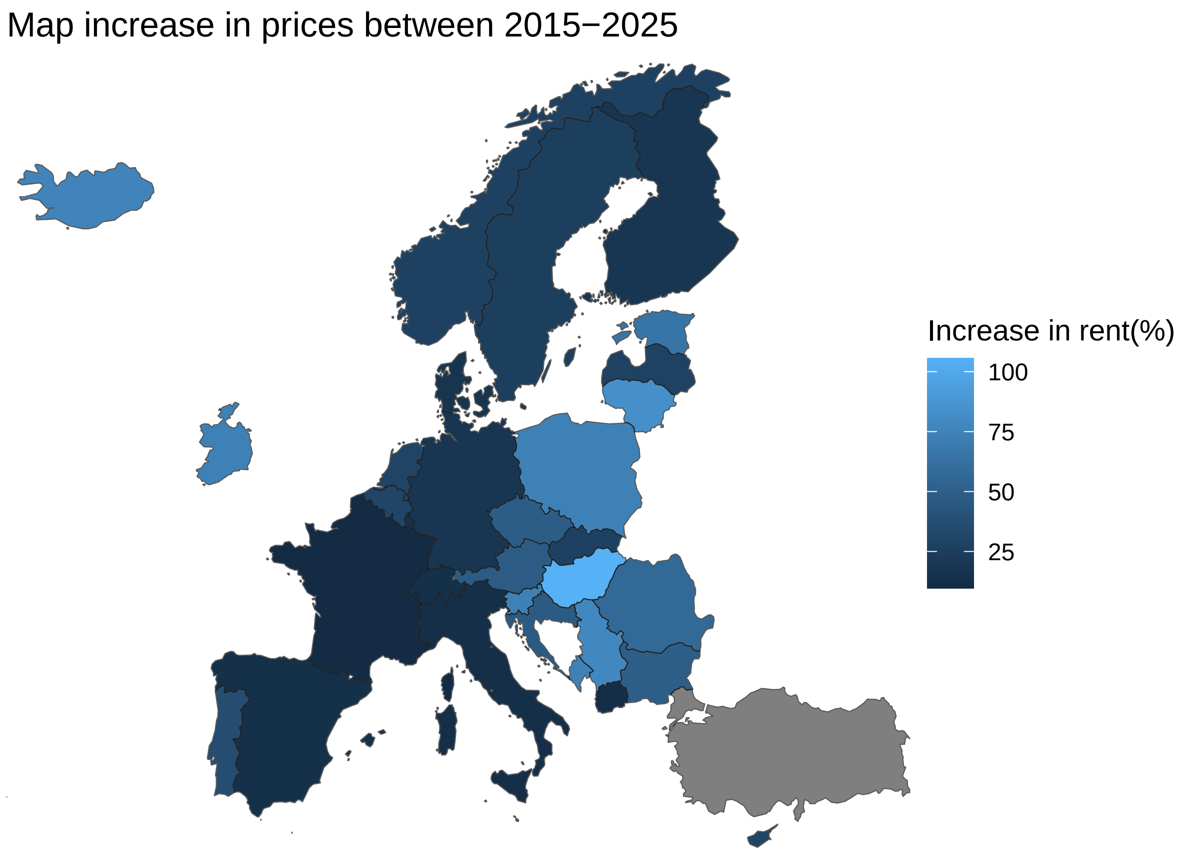
TIME	avg_wage_growth	Var
France	1676.000	9.72
Germany	1849.500	17.90
Netherlands	1918.667	29.27

3.3 Visualize temporal variation



The graph above directly tests our social question. It shows that in 2020 to 2025 the minimum wage growth is growing faster (blue) than the HICP inflation (red). This graph suggests that at the average-level the minimum-wage policy broadly kept up with the cost of living over this time period. Furthermore, it shows that there was most likely a deliberate choice to increase wages after the inflation spikes in 2021-2022. However, keep in mind that this graph shows an average which hides large differences between countries (see sections 3.4 and 3.5).

3.4 Visualize spatial variation



The map shows how much consumer-prices rose in each country between 2015-2025 (not each countries data starts in 2015), and makes clear that not each country has the same growth in prices. With some countries having far steeper growths than others. The largest increases are seen in Eastern-Europe, with a clear example being Hungary that had a growth of 100%. Meanwhile, the countries in Western-Europe, such as France, Germany and Italy saw smaller rises (around 25%). This matters for the social problem, because the countries who face far higher growth in costs of living, will require bigger rises in wage otherwise it will not be sufficient and would leave them off worse. This graph also explains that section 3.3 hides this data, this proves that a single trend line cannot capture each country's situation. (Türkiye is shown in gray since there was no data)

3.5 Visualize sub-population variation

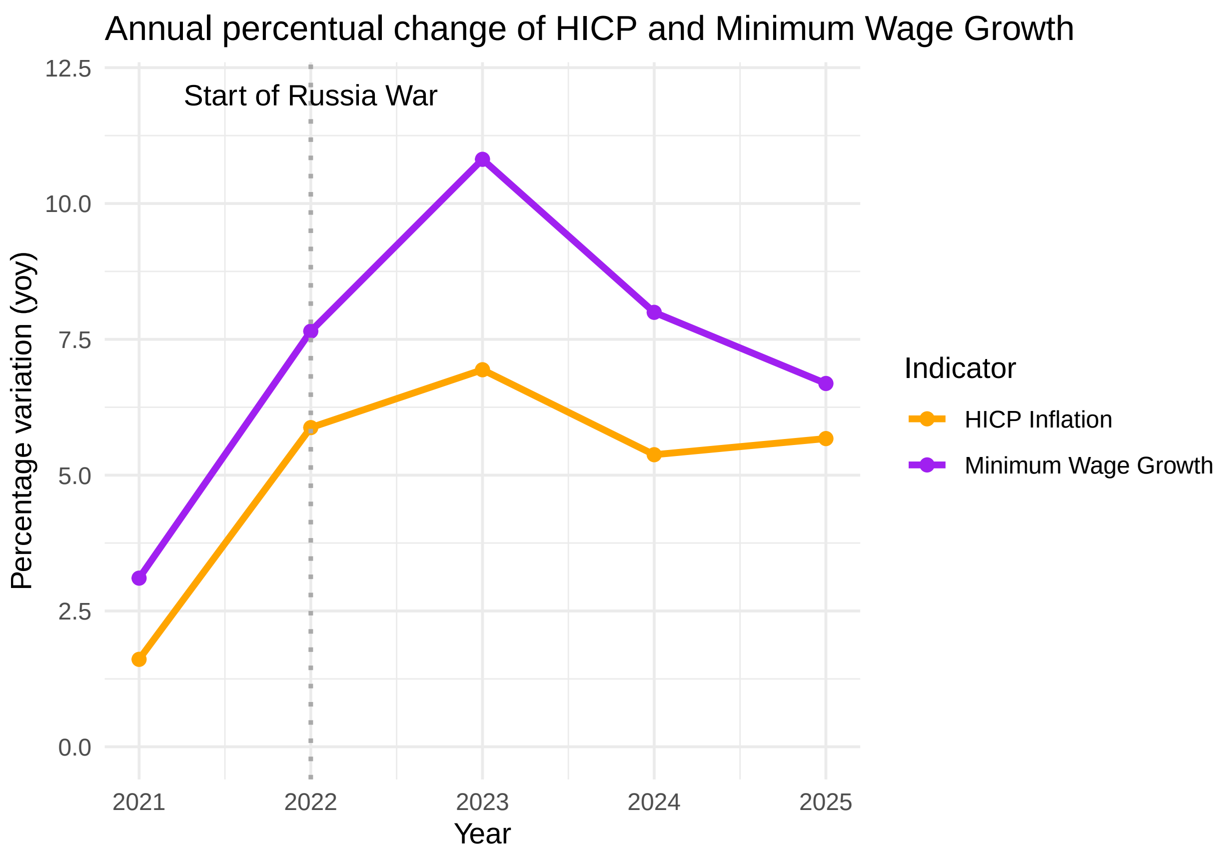
What is the poverty rate by state?

This plot above shows price inflation between countries who have a statutory minimum wage (salmon) and the ones that do not (cyan). Countries with a minimum wage saw both higher and far more variable inflation, a median around 48% and a spread of a 100%, while countries without a minimum wage were lower and tightly clustered together, with a median of 26%. First impression, this looks like having a minimum wage comes with higher inflation, but that is not entirely true. The countries with "no minimum wage" is made up of countries that are wealthy and economically stable (such as Denmark, Sweden, Finland, Austria and Switzerland). Meanwhile the group with minimum wage includes countries from Central Europe and Eastern Europe that have experienced sharp rises in prices. The two group have many differences, not only in minimum wage policies, so this chart shows association, not a cause. For the social problem at hand, it highlights that low-income workers in minimum wage countries often faced far steeper changes in cost-of-living, making it far more important whether their wages kept up.

Two countries in the "no minimum wage" group are outliers, which are Iceland and Austria, showing higher inflation regardless. This shows that the grouping is not always an indicator of whether or not you have high inflation.

3.6 Event analysis

Analyze the relationship between two variables.



Here you provide a description of why the plot above is relevant to your specific social problem. #===
Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ... #===
5.2 Reference list

International Labour Organization. (2022). Global Wage Report 2022–23: The impact of inflation and COVID-19 on wages and purchasing power. ILO. https://www.ilo.org/publications/flagship-reports/global-wage-report-2022-23-impact-inflation-and-covid-19-wages-and

Use APA referencing throughout your document.